

DigitalTwin.ai — Model Parameter Reference

Every input and derived quantity fed to the model, with what it is and why it is there. Table 1 is universal: identical on every line, in every segment, in every plant. Table 2 is assigned per station type, because drift detection transfers across equipment families but fault-mode classification does not.

Table 1 — Bottleneck detection and prediction (universal across all lines)

Parameter	Explanation
RAW INPUT	
unit_id	Identifier of the vehicle or unit. The backbone that links every event to one physical car.
station_id	Which station the event occurred at.
scan_event	Whether the unit entered or exited the station.
timestamp	When the event occurred. Subject to per-controller clock skew, so never assume a common clock.
buffer_id	Which buffer, between which pair of stations.
buffer_level	Number of units currently held in the buffer.
buffer_capacity	Maximum units the buffer can hold. Needed to turn a level into a fill fraction.
variant_id	Product variant of the unit. Different variants load stations differently.
calendar_window	Scheduled break, lunch, changeover or preventive-maintenance period.
station_state	Direct working / blocked / starved / down tag where a PLC exposes it. Optional upgrade that raises provenance from inferred to measured.
PER-STATION (LOCAL)	
dwelt_per_unit	Time the unit spent inside the station. Includes waiting, so it is not processing time.
processing_time_per_unit	Work time only, after blocked and starved time is removed. The honest measure of how long the station takes.
units_completed_per_window	Throughput of that station in the window. Drives the sample size behind every estimate.
active_period_length	Longest uninterrupted span of working-or-down. Roser's criterion: a period is not broken by the station's own repair, only by a neighbour.
active_period_share	Fraction of the window spent inside active periods.
down_time	Time spent in breakdown state within the window.
availability	(scheduled - down) / scheduled. Prices downtime exactly once.
effective_cycle_time	processing / availability. The primary ranking key: a 55 s station at 90% availability effectively needs 61 s per car.
baseline_processing_time	The station's own healthy mean, learned from an in-control window. Drift is judged against self, never against takt.
baseline_spread	The station's own healthy standard deviation.
cusum_deviation	Accumulated evidence that processing time has shifted from its own baseline. Must carry state across windows, not be recomputed each time.
COUPLING (BETWEEN STATIONS)	
blocked_share	Fraction of the window unable to hand off because the downstream station is full. Points to a constraint downstream.
starved_share	Fraction of the window waiting because upstream delivered nothing. Points to a constraint upstream.
forced_idle_share	blocked + starved. The constraint is the station with the least of this.
buffer_fill_fraction_upstream	How full the buffer feeding this station is.
buffer_fill_fraction_downstream	How full the buffer this station feeds is.
buffer_slope_upstream	Rate of change of the upstream buffer level. Whether it is filling matters more than whether it is full.
buffer_slope_downstream	Rate of change of the downstream buffer level.
arrow_direction	Whether this station's blocking exceeds the next station's starvation. Arrows converge on the constraint.
turning_point_flag	The station at which the sign of (blocked - starved) reverses along the line.
inter_departure_interval	Gap between successive units leaving the station.
inter_departure_variance	Smallest at the constraint, because the constraint is the one station neither blocked nor starved.

Parameter	Explanation
arrival_gap	Gap between successive units arriving. The starvation signature.
bottleneck_walk_verdict	Momentary constraint derived from combined process-state and queue-state rules. One of only three methods that finds a momentary rather than average bottleneck.
FORWARD-LOOKING (PREDICTION)	
time_to_buffer_full	(capacity - level) / fill rate. Minutes until the upstream station blocks. Pure arithmetic, works on day one with no history.
time_to_buffer_empty	level / drain rate. Minutes until the downstream station starves.
upcoming_variant_sequence	Variants of the next N units, known in advance from the MES. Free lookahead that requires no forecasting.
predicted_variant_load	Expected extra load per station implied by the upcoming sequence.
time_to_next_break	Breaks are the only mechanism that ever drains a backlog while input continues.
time_to_shift_end	The same drain mechanism at larger scale.
time_to_changeover	A known future disturbance that sweeps down the line station by station.
current_constraint_id	Which station is constraining right now.
constraint_episode_duration	How long the current constraint has held. Feeds the persistence filter.
constraint_transition_history	Sequence of past constraint moves. Fits the shifting-bottleneck transition model.
drift_rate	Seconds of processing time gained per hour. FLAG ONLY - use to say a station is drifting, never to forecast that it will constrain.
CONTEXT AND CONFOUNDERS	
time_since_break	Post-break warm-up genuinely inflates cycle time and will otherwise be read as degradation.
on_break_flag	Break time must be excluded from denominators; it is not the station's fault.
changeover_active_flag	A setup wave sweeping the line is not a bottleneck.
pm_window_flag	Planned maintenance is priced differently from unplanned downtime.
andon_active	An operator has pulled the cord and stopped the line.
andon_reason_code	The only human-assigned explanation for why a stop happened.
rework_reentry_flag	A unit revisiting a station it has already passed. Naive reconstruction double-counts it.
head_supply_available	Whether material is available at the front of the line. Its absence starves everything and hides the real constraint.
META (CARRIED ON EVERY FIELD)	
provenance	measured / inferred / predicted. A measured value and an estimate must never look identical.
confidence	Calibrated against known truth, never asserted. Validate by switching off sensors you do have.
units_in_window	Sample size behind the estimate. Directly drives how much confidence is warranted.

Table 2 — Defect detection (roles fixed; channels assigned per station type)

Parameter	Explanation
CHANNELS (ASSIGN BY ROLE, NOT BY SENSOR NAME)	
primary_spec_channel	The channel carrying the engineering tolerance; it defines what a defect is. Torque at a nutrunner, weld resistance at a spot weld, film thickness in paint.
secondary_channels	Channels that move with real degradation but stay flat under a sensor fault. This is the discriminant: without them you cannot tell a dying tool from a lying sensor.
environmental_channels	Shared signals that move many stations at once - ambient temperature, shop air pressure, humidity. Without them a warm afternoon looks like forty tools failing.
SPECIFICATION AND JUDGEMENT	
target_value	Engineering target for the joint or operation.
spec_low	Lower tolerance limit. Without limits you can detect change but never call a defect.
spec_high	Upper tolerance limit.
controller_verdict	The controller's own OK / NOK decision, held separately from the measured value. The gap between reading and verdict is itself the sensor-fault signal.

Parameter	Explanation
program_id	Tightening program, weld schedule or paint recipe in use. Different programs have different normal behaviour; a change otherwise reads as drift.
IDENTITY AND LINKAGE	
vin	Vehicle identifier. The genealogy backbone that turns an alarm into a containment list.
station_id	Where the operation was performed.
tool_id	Which tool performed it.
tool_serial	The physical unit, so a tool replacement is distinguishable from a tool recovering.
material_lot_id	Fastener, paint or adhesive batch. Enables the cross-tool check that separates a material fault from many tool faults.
timestamp	When. Needed to bound the containment window from drift onset.
INTERVENTION HISTORY	
maintenance_event	Service performed. Produces a step change in the channel that is indistinguishable from a fault unless you ingest it.
calibration_event	Recalibration. Will otherwise be alarmed on every single time.
tool_replacement_event	Resets the degradation clock to zero.
cycles_since_service	Primary covariate for wear. You cannot estimate degradation against an unknown starting point.
OUTCOME (THE DELAYED LABEL)	
eol_test_result	End-of-line pass or fail. Arrives dozens of stations later, which is the entire problem being solved.
inspection_result	Intermediate quality gate outcome.
defect_code	What was actually found to be wrong.
warranty_field_failure	The latest and most expensive label, and the only one that reflects true escapes.
OPERATING CONTEXT	
operating_mode	production / setup / warm-up / test. A reading taken during setup is not a production reading.
shift_id	Shift-level confounder shared across many tools.
time_since_break	Warm-up affects process channels as well as cycle times.
DERIVED	
baseline_mean_per_channel	Healthy mean, per channel per tool, learned from an in-control window.
baseline_spread_per_channel	Healthy standard deviation, per channel per tool.
cusum_state	Accumulated drift evidence per channel. Two numbers of state, microseconds per reading.
ewma_state	Exponentially weighted moving average per channel.
spread_ratio	Change in variability. Bearing wear widens the distribution without moving the mean at all.
primary_to_secondary_shift_ratio	Dimensionless discriminant separating the fault modes. Scale-invariant, so it transfers rather than learning one generator's units.
station_health_score	Aggregate of per-channel evidence, normalised by how many channels were present. Degrades gracefully when a channel is missing.
cross_tool_lot_simultaneity	Whether other tools consuming the same lot alarmed at the same time. Strong common-cause evidence, not proof of causation.
distance_to_spec_limit	How close the current reading sits to the tolerance boundary.
containment_list	The units built on the affected tool since drift onset.
alarm_state	Acknowledge and rearm status. Without it, one latched event masks every later fault on those tools.
META (CARRIED ON EVERY FIELD)	
provenance	measured / inferred / predicted.
confidence	Calibrated, never asserted. Lower automatically when channels are absent.